4.39 The baseline IoH run-off rates are shown on Figure 4f below.

Event	IoH 124 (50ha)	IoH 124 Scaled to 1ha
1 in 1 year (l/s)	186.7	3.73
1 in 100 year (l/s)	700.7	14.01

Figure 4f: IoH124 Baseline Run-off Rates

Hydraulic Design

4.40 Preliminary hydraulic design of the storm water management train has been completed using the WinDES Source Control model for the network described in the preceding paragraphs of this chapter. The drainage system is shown in outline on drawings 1335/DR/02 - 04 contained in Appendix 1.

4.41 The following paragraphs provide details of how each of the three stages combine to reduce the discharge from the proposed development to rates equivalent to the agreed unit discharge rates.

4.42 In essence, each stage of the treatment process provides has an inflow and outflow. Each of the stages are linked in series, with each progressive stage flattening the peak flow resulting from the rainfall progressively to the agreed discharge criteria. The most up to date WinDES system allows cascading detention analysis to be competed to accurately model this behaviour. Figure 4g below illustrates schematically this approach:

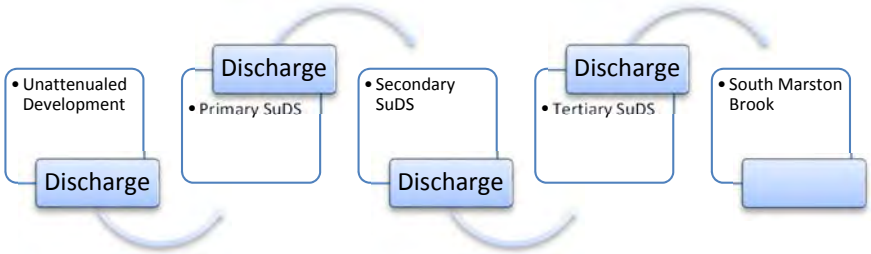

Figure 4g: SuDS Approach

4.43 Best practice methods have been employed by performing detention routing calculations for both the 1 in 1 and 1 in 100 year inlet and outlet return periods using the WinDES Source Control module. Employing the lower and upper end return periods with common characteristics provides for detention that will ensure peak outflows are within the agreed discharges for the full range of storm events. The summary calculations are contained in Appendix 1.

Primary Drainage System

4.44 The characteristics of each new primary sub-catchments have been determined to understand the contributing areas and benefits to be provided by way of the proposed source control measures. Drawing 1335/DR/02 in Appendix 1 shows the layout of these primary sub-catchments.

4.45 Without the benefit of a detailed plot level masterplan, it is not possible to design in detail and appraise the function of the individual source control systems down to plot level. However, previous experience shows that by implementing source control measures such as permeable paving and the like, reduced run-off equivalent to reduced impermeable areas of circa 15% is readily achievable. For the purposes of the primary Drainage System appraisals and inputs to the secondary system, flows from the WinDES Source Control model assessment, at the sub-catchment level, have therefore been modelled as equivalent to a reduced contributing catchment of 15%.

4.46 The calculations demonstrate that by providing the primary SuDS features and thus reducing the impermeable area across a catchment it is possible to reduce discharge rates to the secondary SuDS features. Figure 4h, below summarises the overall reductions for each catchment.

Sub-Catchment	Sub-Catchment Area (Ha)	Impermeable Area (Ha)	1 in 1 run-off (l/s)	1 in 100 run-off (l/s)	Reduced Impermeable Area (Ha)	New 1 in 1 run-off (l/s)	New 1 in 100 run-off (l/s)
I	4.50	2.48	9.23	34.67	2.10	7.85	29.47
II	1.41	0.78	2.89	10.86	0.66	2.46	9.24
III	2.63	1.45	5.40	20.27	1.23	4.59	17.23
IV	1.25	0.69	2.56	9.63	0.58	2.18	8.19
V	1.33	0.73	2.73	10.25	0.62	2.32	8.71
VI	1.58	0.87	3.24	12.17	0.74	2.76	10.35
VII	4.22	2.32	8.66	32.52	1.97	7.36	27.64
VIII	2.81	1.55	5.76	21.65	1.31	4.90	18.40
IX	3.94	2.17	8.08	30.36	1.84	6.87	25.81
X	2.94	1.62	6.03	22.65	1.37	5.13	19.26
XI	3.34	1.84	6.85	25.74	1.56	5.82	21.88
XII	1.98	1.09	4.06	15.26	0.93	3.45	12.97
XIII	2.22	1.00	3.73	14.00	0.85	3.17	11.90
XIV	1.07	0.59	2.20	8.24	0.50	1.87	7.01
XV	1.75	0.96	3.59	13.48	0.82	3.05	11.46
XVI	4.72	2.60	9.68	36.37	2.21	8.23	30.91
XVII	6.06	3.33	12.43	46.70	2.83	10.57	39.69
XVIII	7.84	4.31	16.08	60.41	3.67	13.67	51.35
XIX	2.03	1.12	4.16	15.64	0.95	3.54	13.30
XX	7.47	4.11	15.32	57.56	3.49	13.03	48.93
XXI	0.55	0.30	1.13	4.24	0.26	0.96	3.60
XXII	4.87	2.68	9.99	37.53	2.28	8.49	31.90
Unattenuated Development Discharge			**143.82**	**540.20**			
Reduction						**15%**	**15%**
New Site Discharge (After Primary Reduction)						**122.25**	**459.17**

Figure 4h: Summary run-off & detention assessment output

4.47 The figures above show an estimated 540.2l/s would be generated from the development if no flow attenuation were provided. The 15% reduction due to source control reduces this flow to 459.17l/s.

4.48 The primary system discharge has thereafter been routed through a combined secondary and tertiary SuDS system cascading analysis; this is described in more detail below.

4.49 While a detailed masterplan for the proposed development is not available, Figure 4i below shows a typical arrangement of Primary SuDS measures within a residential parcel.

Figure 4i: Primary Source Control Typical Layout

Secondary Features

4.50 The secondary channels will be designed to provide both storage and conveyance functions. The contributing areas for those of the secondary catchments providing a significant storage purpose have been appraised for input into the WinDES cascading analysis together with the secondary system features. Drawing 1335-DR-03 in Appendix 1 shows the layout of these features and associated catchments.

4.51 The secondary SuDS systems include a number of linear storage and conveyance channels, which are designed to attenuate flows to their maximum available capacity while providing conveyance through the site towards the outfall. Within each of the secondary catchments, the following channel characteristics are proposed:

Sub-Catchment	Channel Length (m)	Channel Volume (m^3)
1	860	3,159
2	200	920

Figure 4j: Secondary Suds Information

Tertiary Features

4.52 As with the Primary and Secondary systems, the Tertiary catchment criteria and treatment systems have been appraised to allow cascading appraisal to be completed. Drawings 1335-DR-04 in Appendix 1 show the illustrative layout of these features and associated catchments.

4.53 Due to the efficiency and capacity within Secondary Catchments 3 and 5, it is considered unnecessary for these to thereafter be passed through a tertiary storage area.

4.54 The calculations predict that the Tertiary Storm Water System will reduce peak discharges from the contributing catchments during the 1 in 100 year plus CC event to 186.5l/s. Detention storage extending to circa 17,144m^3 will be required during the critical 1 in 100 year plus CC event storm.

4.55 Figure 4k, below summarises the overall detention requirements. The summary calculations are contained within Appendix 1.

Sub-Catchment	Impermeable Area (Ha)	1 in 100 run-off (l/s)	Detention volume for 100 Year event (m^3)
A	7.26	24.6	2,061
B	7.05	40.4	4,476
C	4.49	32.1	2,687
D	5.99	29.4	3,945
E	6.03	29.5	3,975
Tertiary SuDS Discharge		**156.0**	
Total Volume			**17,144**

Figure 4k: Summary run-off & detention assessment output

Climate Change Effects on Hydraulic Design

4.56 In accordance with legislative requirements, the detention proposals have also been assessed for the potential effects of climate change. The 1 in 100 year (1% AEP) return events have been modelled for 30% climate change (including peak rainfall intensity). Calculations for the climate change scenarios are contained within Appendix 1.

4.57 Climate change assessments show each detention feature to perform adequately by retaining the additional flows within the system without overflow or unacceptable consequences.

Hydraulic Design Summary

4.58 Preliminary hydraulic design of the storm water management system has been completed for the discharge of flows from the proposed development area to the nearby watercourse network. The calculations show that the SuDS system is able to achieve a peak discharge of 4.83l/s/ha during the 1 in 100 year event. This peak discharge of 4.83l/s/ha represents a reduction in baseline run-off from the site, as shown by IoH124 appraisals, by circa 65.5%.

Scenario	1 in 100 year Run-off (l/s/ha)
Final Site Discharge	4.83
Baseline Discharge	14.01
Overall Reduction	**65.5%**

Figure 4l: Run-off Summary

4.59 The proposed strategic drainage scheme is shown illustratively on drawings 1335-DR-02 - 04 contained in Appendix 1.

4.60 Being an outline planning application, without the benefit of a detailed layout, it is not possible to provide a final drainage layout. However, the development framework plan coupled with the criteria set out in this report will form the framework of the final design at reserved matters stage.

Water Quality

4.61 Impermeable surfaces collect pollutants from a wide variety of sources including cleaning activities, wear from car tyres, vehicle oil and exhaust leaks and general atmospheric deposition (source: CIRIA C609). The implementation of SuDS in development drainage provides a significant benefit in removal of pollutant from development run-off.

4.62 In most cases, contaminants become attached to sediment particles either before entering the water body or upon entry. CIRIA 609 reports that up to 90% of certain contaminants, usually trace elements, are transported in this way leaving a dissolved concentration of circa 10%.

4.63 Many SuDS systems rely on the infiltration of water through the ground layer into permeable sub soils or through sedimentation in low flow storage basins. This settling and filtering of contaminated run off through a fine grained matrix separates the suspended contaminated sediment from the body of water subsequently causing the water to leave the SuDS device in a more polished form than how it entered; porous paving is a prime example of this.

4.64 By implementation of a 'SuDS Treatment Train' it is possible to optimize overall pollutant removal as water will undergo this process of filtering a number of times before being discharged to an appropriate receptor. The overall percentage of removal can be calculated individually for each differing SuDS technique, this is shown by the formula below:

Overall pollutant removal = (TPLxC1) + (RPLxC2) + (RPLxC3) +.....for each other control in series

Where: TPL – Total Pollutant Load
RPL – Remaining Pollutant Load (after previous treatment(s))
C(x) – Suds Control removal efficiency

Figure 4m: Pollutant removal formula as set out in CIRIA C609

4.65 At present, the site and surrounding area that benefits from positive drainage systems do not benefit from any specific measures for storm water quality treatment.

4.66 The SuDS management train being promoted for the development at Swindon EV will be tailored to encourage passive treatment. The three stages of treatment outlined elsewhere in this report will result in very high levels of pollutant load removal as can be demonstrated through the procedures set out in CIRIA C609.

Exceedance Flows

4.67 Careful regard has to be made in respect of potential exceedance flows, being events that are more extreme than current design criteria. Various national guidance has been published on the matter of exceedance flows and measures that should be incorporated into a development to ensure the safety of occupiers and those using the infrastructure.

4.68 The principle aim is to direct any exceedance flows away from properties and along defined corridors. At a local level, this may mean water being conveyed along a length of highway, as long as the predicted flow depths and velocities are acceptable. More strategically, the implementation of conveyance corridors are important in avoiding deep and high velocity flows that present a high risk. The secondary drainage system being promoted for Swindon EV provides a good opportunity to incorporate exceedance flow routes into the design.

4.69 Careful and considered design in other areas, an also reduce the risk. For example the strategic SuDS system being promoted for the development site, provides a layered and disbursed system of treatment across the site, thereby avoiding a traditional and more risky design that might, for example, have all storm water being collected in a strategic spine sewer that conveys flows to a large basin at the bottom of the catchment. This latter system concentrates peak discharges into a single corridor, that if blocked can have unacceptable consequences.

4.70 Other more local measures also should be incorporated. Overflow weirs or the like will be provided on drainage systems, at a level above the 1 in 100 year + 30% flood level, safely to allow more extreme event flows be conveyed away from properties, while at the same time not increasing flood risk to surrounding areas.

4.71 Clearly, many of the measures for dealing with exceedance flows must be dealt with at detailed design stage. However, the strategic layout for Swindon EV provides the framework of a network that can effectively deal with any future exceedance problems.

4.72 The detailed design stage will comply with national standards for protection against exceedance flows.

Implementation & Management Proposals

4.73 The storm water management features will be constructed and operational for each phase in full prior to the first occupation of each phase of new houses across the site.

4.74 It has previously been the case that the functionality of the storm water management system would be ensured by ongoing maintenance, completed by the Local Authority, Drainage Authority, or a private maintenance company as appropriate.

4.75 It was usual for the following maintenance regime to be implemented:

Frequency	Operation
Post major storm events	Inspection and removal of debris.
Every two months	Grass mowing (growing season) & litter removal.
Annual	Weeding & vegetation maintenance. Minor swale clearance. Sweeping of permeable pavements.
2 years	Tree pruning.
5-10 years	Desilting of channels. Remove silt around inlet and outlet structures.
15-20 years	Major vegetation maintenance and watercourse channel works.

Figure 4n: Framework maintenance of detention / retention system

4.76 The Floods and Water Management Act gained royal ascent in April 2010. This confers the responsibility to adopt and maintain the SuDS systems to the Local Authority by requiring SuDS Approving Bodies (SAB's) to be set up within each council.

4.77 The SAB will have a duty to adopt the drainage systems and in accordance with Schedule 3; Para 22 of the Floods and Water Management Act:

22 (1) Where an approving body adopts a drainage system it becomes responsible for maintaining the system.
(2) In maintaining the system the adopting body must comply with national standards for sustainable drainage.

4.78 The SAB will therefore be responsible for developing their framework management plan for maintenance and operation procedures; adjusting the nature of the processes and timing as necessary to ensure the successful operation of the drainage systems.

4.79 At the time of writing, paragraph 22 of the Floods and Water Management Act has yet to be enacted and therefore, the provision for the maintenance of the SuDS features will be agreed with either a public or private management body. During the detailed design stage, a maintenance and management plan will be produced and agreed with the relevant parties to ensure the SuDS features are appropriately maintained.

4.80 The conceptual drainage masterplan proposals outlined in this report will be used for final drainage design and detailing. The storm water management system will be constructed and operational in full prior to the development of each sub catchment discharging into the SuDS feature.

Summary

4.81 A strategy for storm drainage at the site has been developed to meet both national and local policy. The above options outline the viability of the site to employ means of drainage to comply with NPPF, together with the Swindon Borough Council SFRA and other national and local guidance.

4.82 The development drainage system will manage storm water by way of a SuDS management train and ensure peak discharges from the developed land are less than the appraised baseline rates. The system will also provide improvements to the quality of water discharged from the development.

Objectives

4.83 The key objectives for the site drainage will be:

- Implementation of a sustainable drainage scheme in accordance with current national and local policy together with principals of good practice design.
- Development of storm water management proposals that control peak discharges from the site to a rate 56% less than the baseline conditions whilst improving water quality and biodiversity of the site.
- Implementation of the storm water management system prior to first occupation of dwellings coupled with an appropriate maintenance regime.

5 Foul Drainage

Background

5.1 A copy of the Thames Water sewerage network records has been obtained which does not identify any the presence of an adopted foul sewers within the site boundary. Adopted foul sewers service the existing residential and commercial development areas to the west of the proposed development.

5.2 From the Honda factory, foul flows are pumped in a southerly direction towards the A420 Gablecross Junction in a 100mmØ rising main to reach a pumping station on Thornhill Road, south of the rail line.

5.3 Flows are then pumped westerly in another 100mmØ rising main, through the Gablecross junction and along the A420 before making connection with a 150mmØ gravity sewer at Manhole 7507 in the entrance to the Madison Hotel. This 150mmØ gravity sewer directs flows southerly, crossing the A420 and connecting to a pumping station within the Sainsbury's Superstore service area.

5.4 Foul flows are pumped from this location in a 150mmØ rising main in a westerly direction along Oxford Road, entering the residential extent of Swindon before connecting to the 450mmØ gravity foul network at the junction of Oxford Road and Sandgate Mews.

5.5 Approximately 100m west of this point this sewer combines with a number of 150mmØ sewers at Manhole 6103 in the junction of Oxford Road and Winchester Close. At this location, flows are thereafter conveyed in a southerly direction, passing between and number of residential properties and through a number of private gardens in a 400mmØ gravity sewer, reaching manhole 6901 at the cross roads between Brooksby way, Juniper Close and Haven Close.

5.6 From Manhole 6901, a 600mmØ foul sewer conveys flows westerly, passing through a number of private gardens and into Greenbridge Retail and Leisure Park where this 600mmØ sewer combines with a 450mmØ gravity sewer at Manhole 1904 located within the car park area of the Matalan Retail outlet and leaves the estate in a south westerly direction within a 900mmØ sewer.

5.7 Foul water is hereafter conveyed in a westerly direction towards the Rodbourne Sewage Treatment Works, to the west of Swindon, via a network of gravity foul sewers and pumped rising mains serving the residential and commercial areas within Swindon.

5.8 Rodbourne STW currently discharges to the River Ray at Rodbourne. The River Ray is a small watercourse as such, the effluent consent standards are fairly stringent, requiring the treatment process to operate at Best Available Technology (BAT).

5.9 Concerns have previously been raised by both Thames Water and the Environment Agency about the capacity for growth within Swindon. It has been identified there is some capacity in the existing flow consent for additional population, but once the consented flows are reached, any growth will require more stringent treatment than available under BAT, or may result in a deterioration of river quality in the River Ray, and potential failures of statutory water quality targets.

5.10 The currently consent for the Rodbourne STW is shown in Figure 5a below:

Consent	Dry weather flow (m3/d)	Maximum flow to full treatment (m3/d)	Suspended solids (mg/l)	BOD (mg/l)	Ammonia (mg/l)	Phosphorus (mg/l)
CNT.00017	N/A	135,858	17	11	1	1

Figure 5a: Rodbourne STW Consent

Water Cycle Study & Infrastructure Delivery Plan

5.11 As part of the assessments completed by Swindon Borough Council to support the strategic allocation sites and growth options, Halcrow, in conjunction with Thames Water completed a Water Cycle Study in March 2007. The objective of the strategy is to encourage an integrated and sustainable approach to the Water Cycle in Swindon and provide supporting evidence to inform the selection of preferred options for growth.

5.12 The study provided an overview of the foul drainage catchment for the wider Swindon area within the Thames Water catchment looking specifically at the capacity in both the network and treatment systems, along with water efficiency measures to reduce flows from developments and any upgrades required to accommodate for additional flows.

5.13 The WCS has shown that growth in Swindon beyond 2016 cannot be accommodated within the Rodmourne STW if the River Ray is to maintain its current planned water quality. Two options were thereafter mooted within the WCS being:

- **Option 1:** A new STW
- **Option 2:** Upgrading an existing STW

5.14 **Option 1:** Thames Water and the Environment Agency modelled the likely consent standards for a new STW at Seven bridges. The modelling predicted that a new works with a consent set at the most stringent limits applied with BAT can ensure that statutory water quality targets are met in the River Thames. However, there will be some minor deterioration in river quality, as can only be expected with a totally new discharge.

5.15 Coupled with this sewage from the east of Swindon will not naturally gravitate to the new site at Seven bridges and will need to pumped.

5.16 **Option 2:** There is an existing small STW at Wanborough, serving the village of Wanborough which currently discharges to the Liden Brook approximately 3km upstream of its confluence with the River Cole. It is possible that flows currently being treated at Wanborough STW could be transferred to an east of Swindon or Seven Bridges STW.

5.17 More recently, following the production of the Water Cycle Study, Swindon Borough Council produced an Infrastructure Delivery Plan in December 2012 in order to demonstrate that the policies and allocations contained within the Local Plan can be delivered.

5.18 As part of this assessment, Thames water were thereafter contacted to provide an updated position with regards to the provision of foul water treatment capacity for the wider Swindon Area. Correspondence provided by Thames Water on the 2nd March 2012, stated that the intention of Thames Water is to now provide the necessary waste water treatment infrastructure capacity to support forecast growth by upgrading the Rodbourne Sewage Treatment Works.

Design Criteria

5.19 Peak design discharges have been calculated based on the current development criteria as described in Section 2 of this report and for the following:

Residential domestic peak = 4,000 litres / dwelling / day (peak)[6]

5.20 Assessed in accordance with SFA 7th Edition requirements, the development will have a design peak discharge of approximately 111 l/s.

Network Impacts

5.21 Thames Water have been approached to complete a sewer impact study for the proposed development. This study will identify a number of likely points of connection for the proposed development site along with the required infrastructure upgrading.

5.22 The possible points of connection are shown illustratively on Figure 5c below:

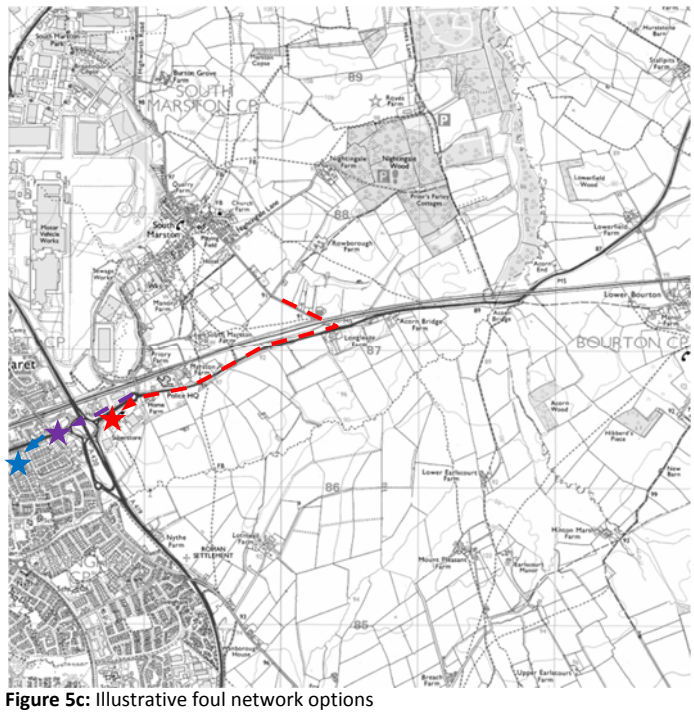

Figure 5c: Illustrative foul network options

Option 1: Foul drainage could be collected within an on-site foul network and directed to a pumping station off the A420 to the south west of the site, adjacent to the existing Sainsburys superstore.

From here foul flows would be pumped in the existing 150mm Ø rising main, in a westerly direction towards Rodbourne Sewage Treatment Works.

Option 2: Foul drainage could be collected within an on-site foul network and pumped in a westerly direction to reach an existing 300mm Ø gravity sewer in Oxford Road.

[6] Sewers for Adoption 7th Edition

A new foul main would be required within from the site along the A420 to reach Manhole 3301 at NGR 418343, 186378.

Option 3: Similarly to Option 2, foul drainage could be collected within an on-site foul network and pumped in a westerly direction to reach the existing 600mm Ø gravity sewer within Juniper Close circa 1km west of the site within the existing residential area of Swindon.
Again, with this solution it would be necessary to implement a number of new foul mains within the existing highway corridors before connecting into manhole 6901 at NGR 417674, 185959.

As with all of the options requiring a connection onto the Thames Water network, this option is subject to final confirmation following the results of the modelling work with regards to the requirements to upgrade the existing foul network to provide capacity.

Implementation Proposals

5.23 The proposed gravity and pumped drainage network across the site will be designed to current Sewers for Adoption 7th Edition Standards, employing the identified points of connection within the Thames Water network. The system will be offered for the adoption of Thames Water Utilities Ltd under S104 of the Water Industry Act 1991.

Summary

5.24 A site drainage strategy has been developed that meets with current regulatory requirements by discharging drainage to a sewerage network with capacity to accommodate the flows.

5.25 Once development is complete, the network conveying flows from the site will be adopted by Thames Water and maintained as part of their statutory duties.

Objectives

5.26 The key development objectives required for the site drainage scheme are:

- Implementation of a drainage scheme to convey water to the local Thames Water network which is designed and maintained to an appropriate standard.

6 Summary

6.1 This FRA has identified no prohibitive engineering constraints in developing the proposed site for the proposed primarily residential usage. The document demonstrates the ability of the proposed development to comply with national, regional and local guidance in the form of NPPF, the Swindon Borough Council SFRA and CIRIA guidance et al along with the agreed scoping report prepared by BCL.

6.2 All built development, with the exception of essential infrastructure, lies in fluvial Flood Zone 1 and hence has a low probability of flood risk.

6.3 Assessment of, ground water and sewer flooding mechanisms does not suggest other than a low risk of flooding at the site. Furthermore, assessment of the overland flow mechanism finds the site to be at very low risk, furthermore and potential mechanism will be adequately mitigated post development.

6.4 A multi-tiered SuDS storm water management system will be employed that results in peak flow discharges from the developed areas of the site being reduced to circa 65.5% below Greenfield rates. Proposals are outlined that show the strategic management system and the FRA outlines additional source control features that will be implemented where appropriate.

6.5 Foul drainage proposals have been developed in consultation with Thames Water that provide for an off-site connection to the existing network. Some off-site upgrading will be necessary to the foul water network.

6.6 The site is able fully to comply with NPPF guidance.

7 Limitations

7.1 The conclusions and recommendations contained herein are limited to those given the general availability of background information and the planned usage of the site.

7.2 Third party information has been used in the preparation of this report, which Brookbanks Consulting Ltd, by necessity assumes is correct at the time of writing. While all reasonable checks have been made on data sources and the accuracy of data, Brookbanks Consulting Ltd accepts no liability for same.

7.3 The benefits of this report are provided solely to the Swindon EDA Consortium for the proposed Phase I Swindon EV development only.

7.4 Brookbanks Consulting Ltd excludes third party rights for the information contained in the report.

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Appendix 1

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1335-SV-01

Notes:

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Rev.	Revision Details	Drawn	Checked	Approved	Date
	PRELIMINARY	PAB			06.12.12
Issue Status		Approved		Date	
Drawn DW	Checked DW	Date	DEC 12		

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Swindon EDA Consortium

Proposed Development
Swindon EV Phase 1

Site Survey

Scale at A1	Drawing No.	Rev.
1:5,000	1335-SV-01	A

0 100 200 300 400 500
METRES

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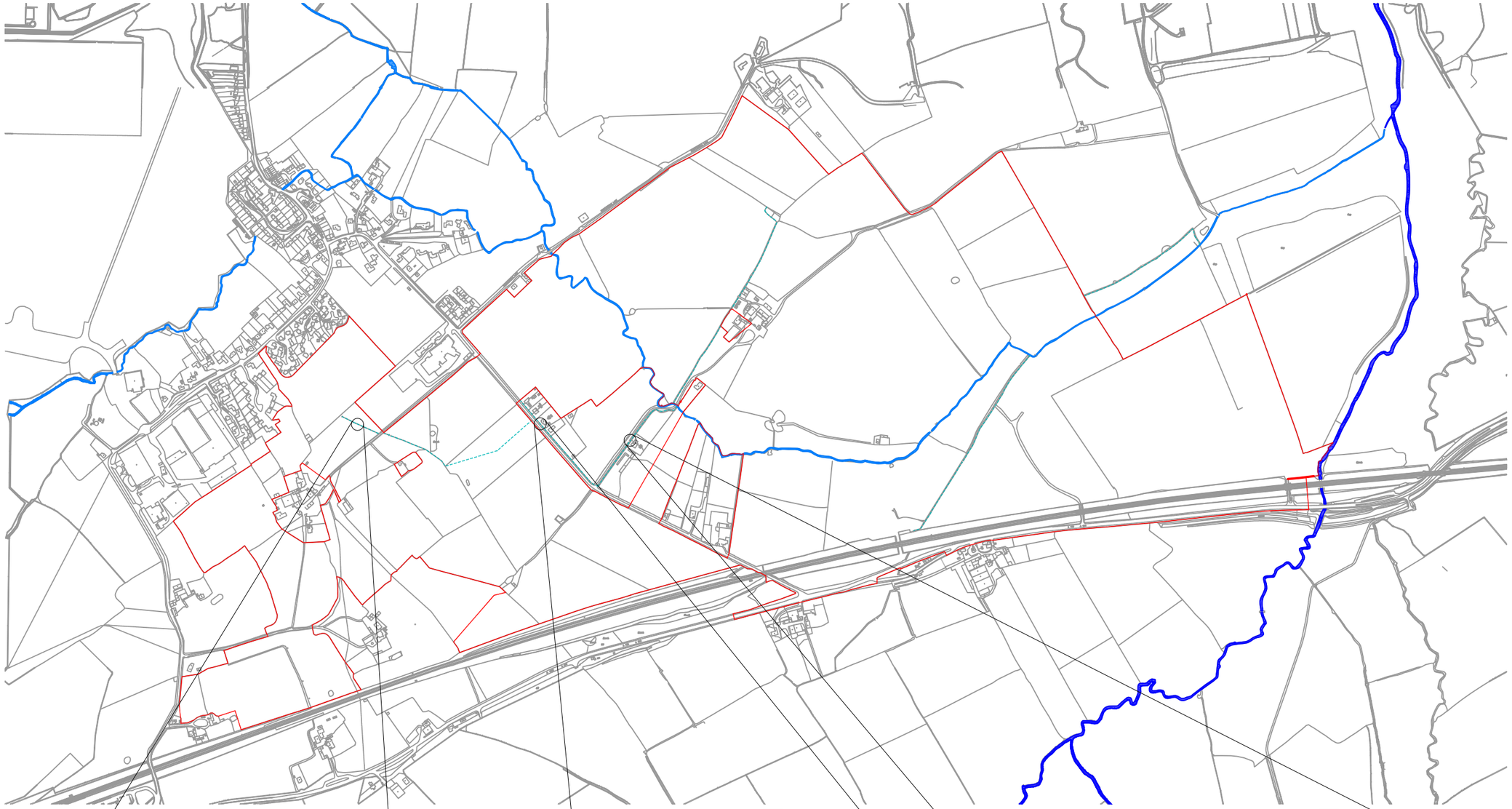
Original Drawing Size A1

1335-DR-01



Notes:

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KEY

- Site Boundary
- River Cole
- South Marston Brook
- Ditchcourse

N

0 100 200 300 400 500
METRES

Rev.	Revision Details	Drawn	Checked	Approved	Date
	PRELIMINARY	PAB			06.12.12
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Drawn CA	Checked DW	Date JAN 12			

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Swindon EDA Consortium

Proposed Development
Swindon EV Phase 1

Drainage Network Plan

Scale at A1	Drawing No.	Rev.
1:5,000	1335-DR-01	A

Original Drawing Size A1

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